

Indian Institute of Technology Guwahati
Probability Theory and Random Processes(MA225)
Problem Set 01

1. Let S be a sample space of a random experiment. Let A , B , and C be three events. What is the event that only A occurs? What is the event that at least two of A , B , C occur? What is the event that both A , B , but not C occur? What is the event of at most one of the A , B , C occurs?
2. Let $S = \{0, 1, 2, \dots\}$ be a sample space. Let $\mathcal{F} = \mathcal{P}(S)$. In each of the following cases, verify if $P(\cdot)$ is a probability.

(a) $P(A) = \sum_{x \in A} \frac{e^{-\lambda} \lambda^x}{x!}, A \in \mathcal{F}, \lambda > 0.$

(b) $P(A) = \sum_{x \in A} p(1-p)^x, A \in \mathcal{F}, 0 < p < 1.$

(c) $P(A) = 0$, if A has a finite number of elements, and $P(A) = 1$, if A has infinite number of elements, $A \in \mathcal{F}$.

3. Let I be any index set. Let $\mathcal{F}_\alpha, \alpha \in I$ be a collection of σ -algebras on $\mathcal{S}$. Prove that $\bigcap_{\alpha \in I} \mathcal{F}_\alpha$ is a σ -algebra.
4. Give a counter-example to show that union of two σ -algebras need not be a sigma algebra.
5. Let $\mathcal{S}$ be the sample space of a random experiment. Let $\mathcal{A}$ be a collection of subsets of $\mathcal{S}$. The smallest σ -algebra containing $\mathcal{A}$ or the σ -algebra generated by $\mathcal{A}$ is defined as

$$\sigma(\mathcal{A}) \doteq \bigcap_{\mathcal{A} \subset \mathcal{F}, \mathcal{F} \text{ is } \sigma\text{-algebra on } \mathcal{S}} \mathcal{F}.$$

- (a) Let A be a non-empty subset of $\mathcal{S}$. Write down the smallest σ -algebra containing A .
- (b) Let A and B be two non-empty subsets of $\mathcal{S}$ such that $A \cup B \neq \mathcal{S}$ and $A \cap B \neq \phi$. Write down the smallest σ -algebra containing A and B .
6. Let $A_1, A_2, \dots, A_n$ be $n > 1$ events. Then prove that

$$P\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n P(A_i).$$

7. Let $A_1, A_2, \dots$ be a sequence of events. Then prove that

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} P(A_i).$$

8. (Principle of inclusion and exclusion) Let $A_1, A_2, \dots, A_n$ be $n > 1$ events. Then prove that

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i) - \sum_{i_1=1}^n \sum_{i_2=1}^n P(A_{i_1} \cap A_{i_2}) + \sum_{i_1=1}^n \sum_{i_2=1}^n \sum_{i_3=1}^n P(A_{i_1} \cap A_{i_2} \cap A_{i_3}) - \dots + (-1)^{n+1} P\left(\bigcap_{i=1}^n A_i\right).$$

9. (Bonferroni's Inequality) Given $n (> 1)$ events $A_1, A_2, \dots, A_n$, prove that

$$\sum_{i=1}^n P(A_i) - \sum_{i=1}^n \sum_{\substack{j=1 \\ i < j}}^n P(A_i \cap A_j) \leq P\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n P(A_i).$$

[Hint: To prove the LHS, use induction.]

10. Let A, B, C , and D be four events such that $P(A) = 0.6, P(B) = 0.5, P(C) = 0.4, P(A \cap B) = 0.3, P(A \cap C) = 0.2, P(B \cap C) = 0.2, P(A \cap B \cap C) = 0.1, P(B \cap D) = P(C \cap D) = 0, P(A \cap D) = 0.1$, and $P(D) = 0.2$. Find
- $P(A \cup B \cup C)$ and $P(A^c \cap B^c \cap C^c)$. (Ans: 0.9 and 0.1)
 - $P((A \cup B) \cap C)$ and $P(A \cup (B \cap C))$. (Ans: 0.3 and 0.7)
 - $P((A^c \cup B^c) \cap C^c)$ and $P((A^c \cap B^c) \cup C^c)$. (Ans: 0.4 and 0.7)
 - $P(D \cap B \cap C)$ and $P(A \cap C \cap D)$. (Ans: 0 and 0)
 - $P(A \cup B \cup D)$ and $P(A \cup B \cup C \cup D)$. (Ans: 0.9 and 1.0)
 - $P((A \cap B) \cup (C \cap D))$. (Ans: 0.3)
11. Let $(\Omega, \mathcal{F}, P)$ be a probability space and let $A, B \in \mathcal{F}$. Show that $P(A \cap B) - P(A)P(B) = P(A)P(B^c) - P(A \cap B^c) = P(A^c)P(B) - P(A^c \cap B) = P((A \cup B)^c) - P(A^c)P(B^c)$.
12. Suppose that $n (\geq 3)$ persons $P_1, \dots, P_n$ are made to stand in a row at random. Find the probability that there are exactly r persons between P_1 and P_2 ; here $r \in \{1, 2, \dots, n-2\}$. (Ans: $2(n-r-1)/(n(n-1))$.)
13. Three numbers a, b , and c are chosen at random and with replacement from the set $\{1, 2, \dots, 6\}$. Find the probability that the quadratic equation $ax^2 + bx + c = 0$ will have real root(s). [Ans: $43/6^3$.]
14. Three numbers are chosen at random and without replacement from the set $\{1, 2, \dots, 50\}$. Find the probability that the chosen numbers are in (a) arithmetic progression, and (b) geometric progression. (Ans: (a) $600/\binom{50}{3}$, (b) $44/\binom{50}{3}$.)
15. A class consisting of four graduate and twelve undergraduate students is randomly divided into four groups of four. What is the probability that each group includes a graduate student? [Ans: $(2 \times 3 \times 4^3)/(15 \times 14 \times 13)$.]
16. Suppose that we have $n (\geq 2)$ letters and corresponding n addressed envelopes. If these letters are inserted at random in n envelopes, find the probability that no letter is inserted into the correct envelop. (Ans: $\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots + (-1)^n \frac{1}{n!}$.)
17. Find the probability that among three random digits there appear exactly two different digits. (Ans: 0.27)
18. Let r indistinguishable balls are placed in n cells numbered 1 through n . Two distributions are said to be distinguishable only if the corresponding n -tuples $(r_1, r_2, \dots, r_n)$ are not identical, where r_i stands for the number of balls in the i th cell.
- Show that the number of distinguishable distributions is $\binom{n+r-1}{r}$.
 - For $r \geq n$, show that the number of distinguishable distributions in which no cell remain empty is $\binom{r-1}{n-1}$.
 - Find the probability that no cell remain empty.

19. A man is given n keys of which only one fits his door. He tries them successively. This procedure may require $1, 2, \dots, n$ trials. Show that each of these n outcomes has probability $1/n$.
20. Suppose that each of n sticks is broken into one long and one short part. The $2n$ parts are arranged into n pairs from which new sticks are formed. Find the probability that the parts will be joined in original order. (Ans: $2^n n!/(2n)!$)